

Elizabeth McDaniel, PhD

Scientist with a passion for bioinformatics, developing open source and reproducible tools, and helping researchers with their analytical challenges.

Email: elizabethmcd93@gmail.com

Website: elizabethmcdaniel.com

GitHub: github.com/elizabethmcd

Skills

Code: R, python, bash, Nextflow, unix, Markdown, SQL

Computing: AWS Batch and EC2, SLURM, HT-Condor, Nextflow Tower, Docker, git/GitHub, Jupyter Lab, Linux systems

Bioinformatics: Genome and metagenome assembly (short and long-read), genome annotation, protein structure analysis, bulk RNA-seq, amplicon sequencing, variant calling

Wet-lab: DNA/RNA extraction, PCR, qPCR, HPLC, lab-scale bioreactor operation

Community Engagement Experience

2023-2024

Arcadia Science internal user group lessons, pre-publication code review

2022-2023

Data Carpentry Genomics Curriculum Advisory Committee Member and Secretary

2019-present

Certified Carpentries Instructor

To-date: Helper/TA for 7 workshops,

Instructor/Course Development for 5 workshops

Co-Chair of the Computational Biology, Ecology, and Evolution (ComBEE) peer study group at UW-Madison

2017-2021

Education

University of Wisconsin - Madison

2016-2021

Ph.D., Microbiology

University of Arkansas

2012-2016

B.S., Biology

Professional and Research Experience

Computational Biologist – *Astera Institute/Microcosm Foods*

Sep. 2024-present

Astera Institute/Microcosm Foods

- Created reproducible computational workflows in Nextflow for predicting bioactive peptides, detecting microbial species in publicly available metagenomic samples, and annotating bacterial genomes with multiple programs.
- Led external outreach and open science strategy, depositing data in FAIR repositories, and working with external platforms for extending the reach and reuse of public datasets.
- Incorporated metagenomic analyses to select appropriate commercial starter culture strains for developing a fermented food product prototype and conducted customer discovery interviews to assess product-market fit for the novel food product.

Computational Scientist - *Arcadia Science*

Oct. 2022-Aug 2024

- Created reproducible computational workflows in Nextflow for both bespoke analyses such as RNA sequence mapping and protein structure analysis and production-ready pipelines for metagenomics and genome assembly on AWS Batch, AWS EC2 instances, and/or Nextflow Tower.
- Assisted scientists with their analytical challenges for pilot projects on quick timescales, provided pre-publication code review, and created internal tools to help scientists understand and visualize their data.
- Communicated the development of open-source tools and scientific results through a non-journal-based publishing platform.

Postdoctoral Researcher - *University of British Columbia, Vancouver, BC*

Nov. 2021-Sep. 2022

- Explored the composition and metabolic activity of microbial communities underlying anaerobic digestion systems using genome-resolved metagenomics, metaproteomics, and stable-isotope probing techniques.

Graduate Research Assistant – *University of Wisconsin – Madison, Madison, WI*

Aug. 2016-Aug. 2021

- Dissertation on the ecology and evolution of enrichment microbial communities simulating enhanced biological phosphorus removal using genome-resolved metagenomics and metatranscriptomics.
- Assisted other researchers with high-throughput mining and annotation of microbial metagenomes.
- Taught workshops on reproducible coding skills with R, python, git/GitHub as a Certified Carpentries Instructor.

Selected Publications

Discovering shared protein structure signatures connected to polyphosphate accumulation in diverse bacteria. Avasthi P., Celebi F.M., McDaniel E.A. *Arcadia Science*. Oct. 2023. DOI: 10.57844/arcadia-ac10-23e7

Diverse electron carriers drive syntrophic interactions in an enriched anaerobic acetate-oxidizing consortium. McDaniel E.A., Scarborough M., Mulat G.M., Lin X., Sampara P.S., Olson H.M., Young R.P., Eder E.K., Attah I.K., Markillie L.M., Hoyt D.W., Lipton M.S., Hallam S.J., Ziels R.M. *ISMEJ*. Oct. 2023. DOI: 10.1038/s41396-023-01542-6.

Quickly preprocessing and profiling microbial community sequencing data with a Nextflow workflow for metagenomics. Dutton R.J., McDaniel E.A. *Arcadia Science*. May 2023. DOI: 10.57844/arcadia-7etp-pj24

TbasCO: Trait-based Comparative 'Omics Identifies Ecosystem-Level and Niche-Differentiating Adaptations of an Engineered Microbiome. McDaniel E.A.#, van Steenbrugge J.J.M.#, Noguera D.R., McMahon K.D., Raaijmakers J.M., Medema M.H., Oyserman B.O. *bioRxiv*. Dec. 2021. DOI: 10.1101/2021.12.04.471239

Expanded Phylogenetic Diversity and Metabolic Flexibility of Mercury-Methylating Microorganisms. McDaniel E.A., Peterson B., Stevens S.L.R., Tran P.Q., Anantharaman K., McMahon K.D. *mSystems*. Aug. 2020. DOI: 10.1128/mSystems.00299-20